

Fluid Flow in Pipe Systems

Group #2

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Introduction

- The objective of this experiment is to obtain various energy loss coefficients such as friction factor, f , loss coefficient, K , and flow coefficient C_f , for different types of pipes and fittings.
- Experimentally obtained values are compared to theoretical values to test them against known correlations for fluid flow in pipes.

Theory

- Pressure drop is recorded and then the friction factor is calculated using equation (1).

$$f = \frac{\Delta P D}{2 L \rho V^2} \quad (1)$$

ΔP = pressure drop

D = diameter

L = length

ρ = density

V = velocity

Theory cont.

- The head loss h_L , in other words the pressure drop, refers to the measurement of the energy displacement in a system due to friction and is calculated using equation (2).

$$h_L = \frac{\Delta P}{\rho} = f \frac{L}{D} \frac{V^2}{2} \quad (2)$$

Theory cont.

- When the flow of the fluid is laminar the resistance is due to viscous forces and is not dependent on the roughness of the pipe being used. The friction factor is then calculated using equation (3).

$$f = \frac{64}{Re} \quad (3)$$

Re = Reynolds number

Theory cont.

- The friction factor of turbulent flow is empirically calculated using the Colebrook equation (4).

$$\frac{1}{\sqrt{f}} = -2 \log \left(\frac{k_s}{3.7 D} + \frac{2.51}{Re \sqrt{f}} \right) \quad (4)$$

where k_s is the sand roughness of the pipe

Theory cont.

- The head loss due to most pipe fitting can be calculated using equation (5).

$$h_L = K \frac{V^2}{2} \quad (5)$$

- K is defined by equation (6).

$$K = f \frac{L}{D} \quad (6)$$

Theory Cont.

- To calculate the orifice head loss, Bernoulli's equation (7) can be used to describe the flow of the fluid as long as the speed of the fluid is sufficiently subsonic.

$$\Delta P = P_1 - P_2 = \frac{1}{2}\rho V_2^2 - \frac{1}{2}\rho V_1^2 \quad (7)$$

Theory cont.

- The velocity term can be replaced by the cross-sectional areas and the flow can be replaced by the volumetric flowrate Q , resulting in the equation (8).

$$\Delta P = \frac{1}{2} \rho Q^2 \frac{1}{A_2^2} \left[1 - \left(\frac{A_2}{A_1} \right)^2 \right] \quad (8)$$

Theory cont.

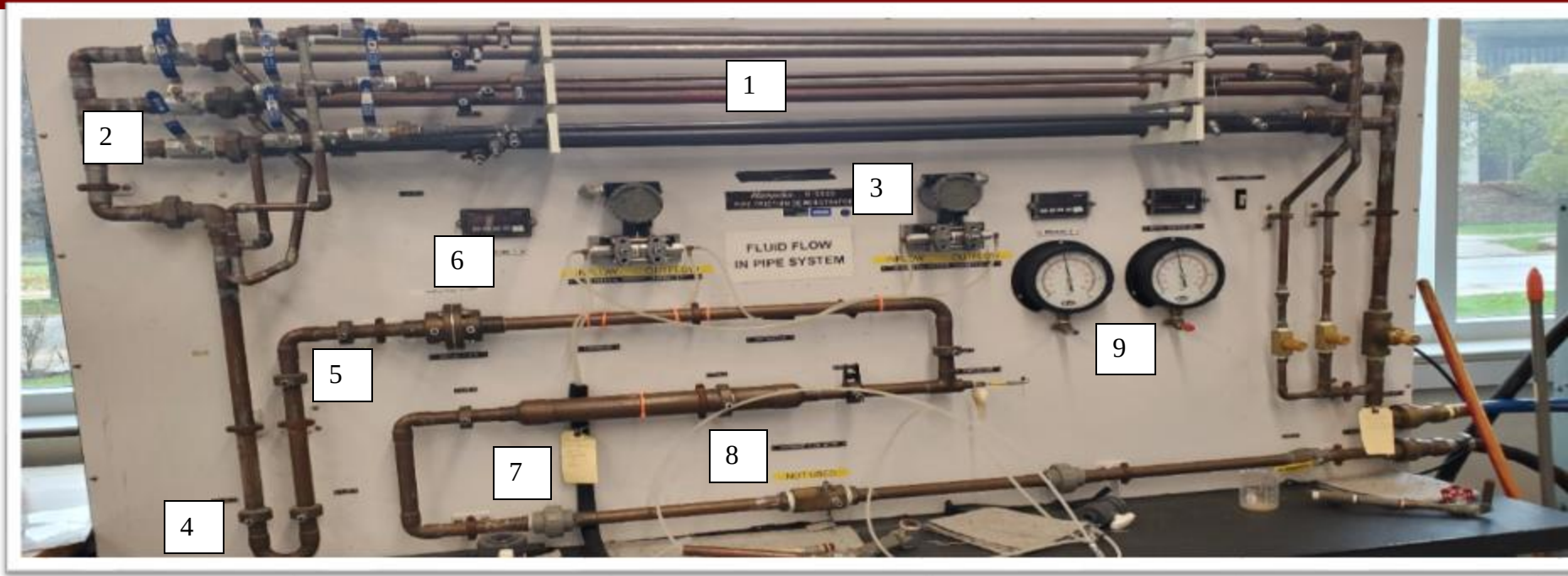
- With some rearranging Q is found to be:

$$Q = \sqrt{\frac{2 \Delta P}{\rho}} \frac{A_2}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}} \quad (9)$$

- Applying the discharge coefficient and the flow coefficients Q becomes:

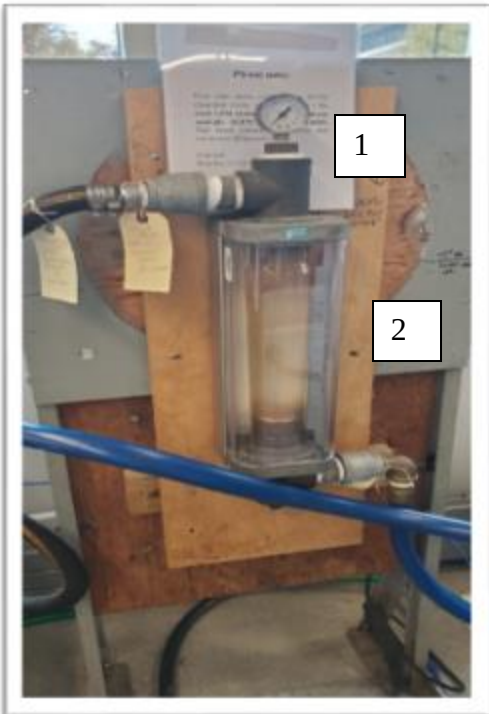
$$Q = C_f A_o \sqrt{\frac{2 \Delta P}{\rho}} \quad (12)$$

Apparatus

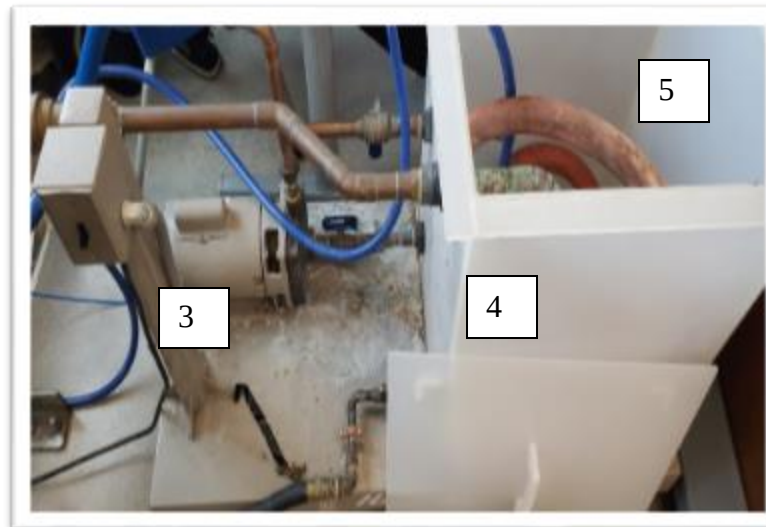


1. Pipe Fittings: Copper, SS, and PVC with Different Diameters
2. Pipe Shutoff Valve: Introduces Water Flow Inside
3. Pressure Transmitter: Measures the Pressure Drop in Pipe
4. U-tube Pipe
5. L-Tube Pipe
6. Orifice Meter
7. Expansion Pipe
8. Contraction Pipe
9. Pressure Gauge: Measures Pressure in the System

Apparatus Continued...



1. Water Pressure Gauge
2. Rotameter: Displays the flowrate of the Water
3. Power to pump: Turns the pump on and off
4. Reservoir Tank: Water is pumped into here for use into the whole system
5. Water Hose: Allows water into the reservoir tank



Material & Supplies

- Graduated Cylinder
- Water
- Timer

Procedure

- Pump into the white reservoir tank
- Rotameter Calibration
 - Adjust the flowrate on the rotameter
 - Collect a volume of water at a controlled time.
 - Do this for 3-4 different flowrates on the rotameter
- Collecting Data from the Pipes
 - Open the main valve and switch the power on
 - Make sure the pressure transmitter is reading zero

Procedure Continued...

- Attach Pressure taps into the pressure sensing ports and then to the ends of the pipe
- Open the Pipe Shutoff valve and adjust the flowrate to the desired value
- Record the pressure and the temperature
- Collect 3-4 different flowrate measurements for each of the pipes.
 - For the smaller sizes, use smaller flowrates
- After finished with one pipe, close the shutoff valve and drain the pressure transmitters

Procedure Continued...

- Measure the length of the straight pipes and the diameters of the U-pipe, L-pipe, 90° pipe, contraction pipe, and the expansion pipe.

Experimental Challenges

- When connecting the pressure transmitters to the bended pipes, make sure the data is making sense or else the pressure transmitters aren't in the correct spots
- For the smaller straight pipes, it's better to have flowrates starting from 10,15, 20 because the flowrate will not go any higher than around 20

Safety

- Watch out for possible exposed wires near the pump and any mechanical hazards if the pump malfunctions
- Make sure the pressure isn't too high, since it can lead to the pipe bursting

Comment Summary

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i. Technical details